

# PESO Quezon City — Labor Rights AI Chatbot

## System Documentation

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## 1. System Overview

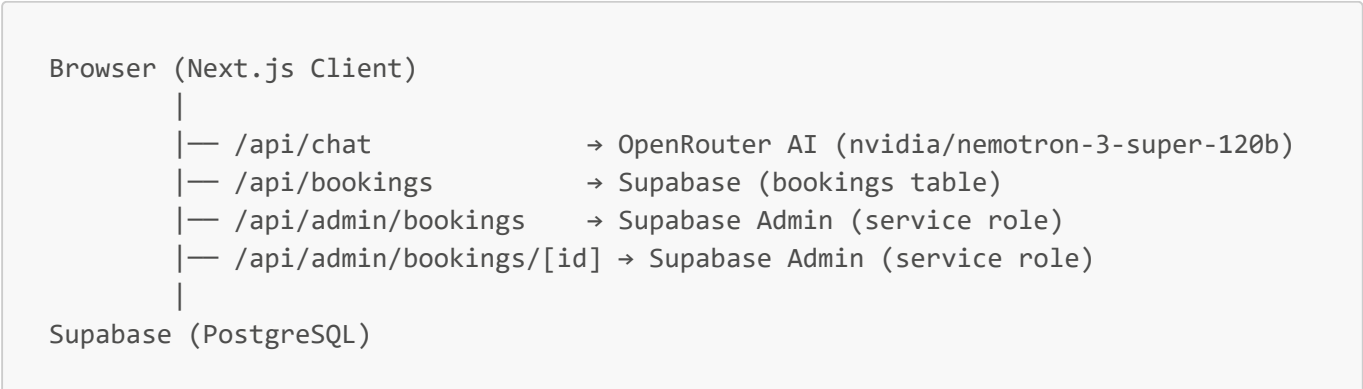
The PESO QC Labor Rights AI Chatbot is a web application built for the **Public Employment Service Office of Quezon City**. It provides workers — including regular employees, contractual workers, kasambahay (domestic workers), and fresh graduates — with accessible information about their rights under the **Philippine Labor Code**.

The system allows users to:

- Ask labor rights questions in Filipino, English, or Taglish
- Compute pay benefits (overtime, 13th month pay, night shift differential)
- Book consultation appointments with PESO QC staff
- Export conversation transcripts

PESO QC staff can manage and track bookings through a password-protected admin dashboard.

## 2. Architecture



└─ bookings table

└─ chat\_sessions table

Stack:

| Layer              | Technology                                          |
|--------------------|-----------------------------------------------------|
| Framework          | Next.js 16 (App Router)                             |
| Language           | TypeScript                                          |
| Styling            | Tailwind CSS v4                                     |
| AI Provider        | OpenRouter (nvidia/nemotron-3-super-120b-a12b:free) |
| Database           | Supabase (PostgreSQL)                               |
| Markdown Rendering | react-markdown                                      |

3. Project Structure

peso-labor-chatbot/

└─ app/

└─ layout.tsx

└─ page.tsx

└─ globals.css

└─ admin/

└─ page.tsx

└─ api/

└─ chat/

└─ route.ts

└─ bookings/

└─ route.ts

└─ admin/

└─ bookings/

└─ route.ts

└─ [id]/

└─ route.ts

└─ components/

└─ ChatWindow.tsx

└─ BookingModal.tsx

└─ lib/

└─ supabase.ts

└─ supabaseAdmin.ts

└─ rateLimit.ts

# Root layout, fonts, metadata

# Main chat page (header + footer)

# Global styles

# Admin dashboard (password-protected)

# AI chat endpoint

# Public booking submission endpoint

# Admin: list all bookings (GET)

# Admin: update booking status (PATCH)

# Main chat UI (tabs, messages, calculator)

# Appointment booking form modal

# Supabase anon client (public)

# Supabase service role client (server-only)

# In-memory IP-based rate limiter

4. Features

4.1 Chat

- Powered by an AI model via OpenRouter
- Responds in the same language the user writes in (Filipino, English, or Taglish)
- Cites specific Labor Code articles and laws in every response
- Generates 2–3 follow-up question chips after each answer
- Conversation history is persisted per browser session via Supabase (`chat_sessions` table)
- Users can copy any assistant message
- Users can rate responses (thumbs up / thumbs down)
- Users can export the full conversation as a `.txt` file
- Users can clear their conversation (also deletes records from Supabase)

## 4.2 Pay Calculator

An offline (no AI) calculator for three common computations:

| Calculator               | Formula                                                                   | Legal Basis        |
|--------------------------|---------------------------------------------------------------------------|--------------------|
| Overtime Pay             | $(\text{Daily Rate} \div 8) \times \text{multiplier} \times \text{hours}$ | Labor Code Art. 87 |
| 13th Month Pay           | $(\text{Monthly Salary} \times \text{Months Worked}) \div 12$             | PD 851             |
| Night Shift Differential | $\text{Hourly Rate} \times 10\% \times \text{NSD hours}$                  | Labor Code Art. 86 |

Overtime multipliers:

- Regular day: **+25%** ( $\times 1.25$ )
- Rest day or holiday: **+30%** ( $\times 1.30$ )

## 4.3 Appointment Booking

Users can submit an appointment request through a modal form with the following fields:

| Field          | Validation                                                                |
|----------------|---------------------------------------------------------------------------|
| Full Name      | Minimum 2 characters                                                      |
| Contact Number | Philippine format: <code>09XXXXXXXXX</code> or <code>+639XXXXXXXXX</code> |
| Preferred Date | Must be a future weekday (Mon–Fri)                                        |
| Concern        | Minimum 10 characters                                                     |

Validation is applied on both the client (inline, on blur) and the server (API route). Submissions are saved to the `bookings` table in Supabase.

## 4.4 Topics Browser

A browsable index of common labor rights topics organized into five categories:

- Sahod at Benepisyo (Wages and Benefits)
- Leaves at Pahinga (Leaves and Rest)
- Dismissal at Kontrata (Dismissal and Contracts)
- Kontribusyon (SSS, PhilHealth, Pag-IBIG)
- Kasambahay (Domestic Workers)

Clicking any question sends it directly to the chat.

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## 5. API Reference

### POST `/api/chat`

Sends a message to the AI and returns a response.

**Rate limit:** 10 requests per minute per IP.

**Request body:**

```
{
  "messages": [
    { "role": "user", "content": "Magkano ang minimum wage?" }
  ]
}
```

**Response:**

```
{
  "message": "Ang minimum wage sa NCR ay ₱695 bawat araw..."
}
```

**Error responses:**

| Status | Meaning                                        |
|--------|------------------------------------------------|
| 400    | Invalid or missing <code>messages</code> array |
| 429    | Rate limit exceeded                            |
| 500    | AI provider error                              |

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### POST `/api/bookings`

Submits an appointment booking request.

**Rate limit:** 3 requests per minute per IP.

**Request body:**

```
{
  "name": "Juan dela Cruz",
  "contact": "09171234567",
  "preferredDate": "2026-05-15",
}
```

```
{  "concern": "Hindi ako binibigyan ng 13th month pay ng employer ko."}
```

**Response:**

```
{ "success": true }
```

**Error response:**

```
{ "error": "Error message here.", "field": "contact" }
```

The **field** property identifies which form field caused the error (**name**, **contact**, **date**, **concern**, or **general**).

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**GET /api/admin/bookings**

Returns all bookings. Requires admin password.

**Authorization:** Bearer <ADMIN\_PASSWORD> header required.

**Response:**

```
{  "bookings": [    {      "id": "uuid",      "name": "Juan dela Cruz",      "contact": "09171234567",      "preferred_date": "2026-05-15",      "concern": "...",      "status": "pending",      "created_at": "2026-05-07T10:00:00Z"    }  ]}
```

**PATCH /api/admin/bookings/[id]**

Updates the status of a booking. Requires admin password.

**Authorization:** Bearer <ADMIN\_PASSWORD> header required.

**Request body:**

```
{ "status": "confirmed" }
```

Valid status values: `pending`, `confirmed`, `done`.

### Response:

```
{ "success": true }
```

---

## 6. Database Schema

### bookings

| Column                      | Type       | Notes                                                                |
|-----------------------------|------------|----------------------------------------------------------------------|
| <code>id</code>             | uuid       | Primary key, auto-generated                                          |
| <code>name</code>           | text       | Full name of the applicant                                           |
| <code>contact</code>        | text       | Philippine mobile number                                             |
| <code>preferred_date</code> | text       | Format: YYYY-MM-DD                                                   |
| <code>concern</code>        | text       | Description of the labor concern                                     |
| <code>status</code>         | text       | <code>pending</code> , <code>confirmed</code> , or <code>done</code> |
| <code>created_at</code>     | timestampz | Auto-set on insert                                                   |

### chat\_sessions

| Column                  | Type       | Notes                                                                              |
|-------------------------|------------|------------------------------------------------------------------------------------|
| <code>id</code>         | uuid       | Primary key, auto-generated                                                        |
| <code>session_id</code> | text       | UUID generated per browser session                                                 |
| <code>role</code>       | text       | <code>user</code> or <code>assistant</code>                                        |
| <code>content</code>    | text       | Raw message content (includes <code>FOLLOW_UP:</code> line for assistant messages) |
| <code>created_at</code> | timestampz | Auto-set on insert, used for ordering                                              |

---

## 7. Environment Variables

Create a `.env.local` file in the project root with the following variables:

```
# OpenRouter API key for AI chat
OPENROUTER_API_KEY=your_openrouter_api_key
```

```
# Supabase project URL (public)
NEXT_PUBLIC_SUPABASE_URL=https://your-project.supabase.co

# Supabase anon key (public, safe for browser)
NEXT_PUBLIC_SUPABASE_ANON_KEY=your_anon_key

# Supabase service role key (private, server-only – never expose to browser)
SUPABASE_SERVICE_ROLE_KEY=your_service_role_key

# Admin dashboard password
ADMIN_PASSWORD=your_admin_password
```

**Security note:** `SUPABASE_SERVICE_ROLE_KEY` and `ADMIN_PASSWORD` must never be prefixed with `NEXT_PUBLIC_`. They are only accessible server-side.

## 8. Labor Data Sources

All labor figures are injected into the AI system prompt. The AI is explicitly instructed to use these values instead of its own training data.

| Data Point                                 | Value                        | Source                                       |
|--------------------------------------------|------------------------------|----------------------------------------------|
| Minimum Wage (NCR, Non-Agriculture)        | ₱695/day                     | Wage Order No. NCR-26, effective May 1, 2026 |
| Minimum Wage (NCR, Agriculture)            | ₱695/day                     | Wage Order No. NCR-26                        |
| Minimum Wage (Retail/Service, ≤15 workers) | ₱695/day                     | Wage Order No. NCR-26                        |
| 13th Month Pay                             | 1/12 of annual basic salary  | Presidential Decree 851                      |
| Night Shift Differential                   | +10% for work 10PM–6AM       | Labor Code Article 86                        |
| Overtime — Regular Day                     | +25%                         | Labor Code Article 87                        |
| Overtime — Rest Day / Holiday              | +30%                         | Labor Code Article 87                        |
| Regular Holiday Pay (not worked)           | 100% of daily rate           | Labor Code Article 94                        |
| Regular Holiday Pay (worked)               | 200% of daily rate           | Labor Code Article 94                        |
| Special Non-Working Holiday (worked)       | +30% premium                 | Labor Code Article 94                        |
| Service Incentive Leave                    | 5 paid days/year             | Labor Code Article 95                        |
| SSS Contribution Rate                      | 15% of MSC (EE: 5%, ER: 10%) | 2025 SSS schedule                            |

| Data Point                        | Value                       | Source               |
|-----------------------------------|-----------------------------|----------------------|
| SSS Monthly Salary Credit Range   | ₱5,000 – ₱35,000            | 2025 SSS schedule    |
| PhilHealth Contribution Rate      | 5% of basic monthly salary  | 2025 PhilHealth rate |
| PhilHealth Income Floor / Ceiling | ₱10,000 / ₱100,000          | 2025 PhilHealth rate |
| Pag-IBIG Contribution             | 2% EE + 2% ER (min ₱200/mo) | 2025 HDMF schedule   |

To update any figure, edit the `SYSTEM_PROMPT` constant in [app/api/chat/route.ts](#).

## 9. Rate Limiting

Rate limiting is implemented in `lib/rateLimit.ts` using an **in-memory store** (a `Map`) keyed by `IP:pathname`.

| Endpoint                   | Limit                      |
|----------------------------|----------------------------|
| <code>/api/chat</code>     | 10 requests per 60 seconds |
| <code>/api/bookings</code> | 3 requests per 60 seconds  |

The IP address is read from the `x-forwarded-for` header (first value) or `x-real-ip`. If neither is present, the key falls back to `"unknown"`.

**Important limitation:** Because the store is in-memory, rate limit counters reset whenever the server process restarts and are not shared across multiple server instances (e.g., in a horizontally scaled deployment). For production at scale, replace the `Map` with a Redis-backed solution.

## 10. Admin Dashboard

The admin dashboard is accessible at `/admin`.

### Login

Authentication is a simple password check. The entered password is sent as a `Bearer` token in the `Authorization` header of the GET request to `/api/admin/bookings`. If the password matches `ADMIN_PASSWORD`, the bookings are returned and the user is considered authenticated. The password is held only in component state — it is never stored in `localStorage` or cookies.

### Booking Management

- All bookings are listed in a table sorted by submission date (newest first)
- Bookings can be filtered by status: All, Pending, Confirmed, Done
- Status can be updated inline via a dropdown — changes are sent to `PATCH /api/admin/bookings/[id]` immediately
- If a status update fails, an error banner is shown at the top of the dashboard



## Logout

Clicking Logout clears the password and bookings from component state. No server-side session is invalidated (stateless auth).

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## 11. Known Limitations

| Limitation                         | Details                                                                                               |
|------------------------------------|-------------------------------------------------------------------------------------------------------|
| In-memory rate limiter             | Resets on server restart; not suitable for multi-instance deployments                                 |
| Admin auth is stateless            | Password is held in React state only; no session expiry or token revocation                           |
| AI response time                   | The free-tier AI model ( <code>nvidia/nemotron-3-super-120b-a12b:free</code> ) can be slow under load |
| No email notifications             | Booking confirmation is on-screen only; staff must check the admin dashboard manually                 |
| Feedback not persisted             | Thumbs up/down ratings are stored in React state only and lost on page refresh                        |
| Chat session scoped to browser tab | <code>sessionStorage</code> is used for the session ID, so each tab has a separate conversation       |